MPhil Thesis: Introduction

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1 Introduction

Pharmaceutical innovation is central towards long-run improvements in health and welfare. New drugs can improve lifestyle quality while changing the treatment possibilities which are available for physicians to administer on patients. At the same time, pharmaceutical innovation is sensitive to policy. The development of a new drug requires a large upfront investment with a long development horizon including patent applications and clinical trials, with a high probability of failure. Since the returns to a successful drug accrue over multiple years and depend on both prices and the volumes that prevail within the market, the incentives to undertake the investment to develop a drug are shaped not only by opportunity, but also by the economic and institutional environment present which determine the expected future profits once the drug reaches the market.

This makes the pharmaceutical market within the United States of America an ideal setting for studying how public policy shapes scientific opportunity. Federal and state governments are now responsible for X% of all prescription drug spending (SOURCE), both as direct purchasers through programmes such as Medicaid and Medicare, and also indirectly through the regulation that governs private insurance. Medicaid is especially important in this respect - it is a large public buyer, but it does not operate as a standard private insurer. Manufacturers that wish to have Medicaid coverage must participate in the Medicaid Drug rebate Program, and states can use Preferred Drug Lists (PDLs) in order to negotiate additional rebates in exchange for preferred formulary placement. Medicaid demand therefore passes through a public procurement system rather than a conventional insurance market, and its institutional features are described in detail in Section 2.

This thesis examines the consequences of one of such reforms for pharmaceutical innovation - the 2014 Medicaid expansion under the the Patient Protection and Affordable Care Act (ACA). The reform produced a well-documented expansion in the size of the Medicaid market, with population coverage increasing by close to twenty million enrollees in the years following the policy change (SOURCE). Under standard market-sized frameworks in the pharmaceutical sector this should increase innovation incentives for drug-classes more exposed to Medicaid demand (Finkelstein, 2004; Dubois et al., 2015). If firms expect to sell more units after a successful round of innovation, the expected returns to R&D rise. However, the additional demand entered the market through Medicaid's procurement institutions rather than through standard private channels, so an expansion of

Medicaid was not simply a positive demand shock for pharmaceutical firms.

The Medicaid setting therefore complicates the prediction under standard pharmaceutical market-sized frameworks. The ACA did not only expand the number of patients under Medicaid, but it also raised the statutory rebate percentages and extended rebate obligations to Medicaid managed care. Therefore, the reform increased the size of the public market while also increasing the share of revenue back to the state. Additionally, as Medicaid demand grows, preferred placement on a state PDL becomes more valuable. States can use this to negotiate deeper rebates from manufacturers. The same policy may therefore generate two opposing effects. On the one hand, Medicaid expansion raises demand. On the other hand, it may increase pricing pressure and reduce the surplus that firms can capture from that demand.

This thesis therefore raises the question of whether the ACA-induced Medicaid expansion raised pharmaceutical innovation, and whether the response dependent on a firm's exposure to the Medicaid procurement institutions. Put simply, was the expected positive relationship between market size and innovation weakened when the additional demand flows through a state-wide monopsony. Many current drug-pricing reforms, including recent Medicare negotiation policies, rely on the same basic trade-off: lowering public spending and consumer prices may increase static welfare, but it may also affect the expected returns to future innovation, raising the relevance of this question

This thesis makes several contributions. First, it constructs a firm-by-therapeutic-class panel linking patent records, FDA approvals, Medicaid prescription data, and PDL placement over 2010-2025. This exercise is non-trivial since the underlying sources do not share common firm, drug, or therapeutic-class identifiers. Second, it studies the innovation response to ACA-induced Medicaid expansion using patents and FDA approvals rather than clinical trial activity, while extending the post-expansion period through to 2025. Third, it moves beyond aggregate therapeutic-class exposure by linking innovation outcomes to firm-level exposure to Medicaid's PDL, allowing the analysis to separate Medicaid demand exposure from PDL exposure and to examine whether responses vary across therapeutic areas. Fourth, it allows the innovation response to vary across ATC therapeutic areas, rather than imposing a single average response across all drug classes. Finally, the thesis conducts PDL counterfactual analysis and builds a theoretical model of firm innovation behaviour to interpret the empirical results and show how an expansion in Medicaid demand can be offset by the Medicaid monopsony wedge. This builds on Garthwaite et al. (2021), who study clinical trial activity around the Medicaid expansion. Compared to their work, this thesis extends the outcome set, time horizon, and level of analysis.

The main empirical finding of this thesis is that therapeutic classes with greater pre-expansion Medicaid exposure experienced a decline in patenting following the 2014 ACA Medicaid expansion, while the evidence for FDA approvals appears to be inconsistent. This suggests that the innovation response to the policy was concentrated earlier in the R&D pipeline rather than at the final point of regulatory approval. Additional specifications reveal that this aggregate response masks heterogeneity, where firms with greater exposure to Medicaid's PDL institutions experience stronger post-expansion reductions in innovation, and the effects differ across therapeutic areas. The negative patenting response is most pronounced in classes where Medicaid exposure is high and products are more substitutable, while other classes show weaker or more positive responses. The theoretical model provides an interpretation for these patterns: while the Medicaid expansion raised expected demand, it also increased a procurement wedge faced by firms through rebates and bargaining. If the procurement effect is large enough, then even while Medicaid demand increases, firms may not expect to earn sufficient profits for innovation incentives to rise. This lends an explanation for why the standard market-size prediction may fail in this setting.

1.1 Medicaid Expansion and Pharmaceutical Innovation

The institutional features described change how the standard market-size prediction should be applied to Medicaid expansion. Under the standard framework, an increase in expected demand raises the expected return to successful innovation, and as a result, firms should possess stronger incentives to invest in therapeutic classes where the ACA increased expected prescriptions. This is the market-size channel documented in the pharmaceutical innovation literature (Finkelstein, 2004; Dubois et al., 2015).

In this setting, however, the relevant quantity is not Medicaid demand but the surplus firms expect to gain from that demand. Medicaid prescriptions are subject to statutory rebates, and gaining preferred placement on a state PDL can require additional price concessions through supplemental rebates. Therefore, the expansion raised expected prescription volume while also increasing the importance attached to the institutions determining what proportion of said revenue that firms can retain. The net effect on innovation is therefore ambiguous: it may be positive if the demand effect dominates, but negative if procurement institutions reduce expected returns sufficiently.

This thesis studies that ambiguity using patents and FDA approvals as measures of pharmaceutical innovation. Patents capture earlier-stage innovation activity, while FDA approvals capture realised innovative output. This allows the analysis to examine whether Medicaid expansion affected different points of the innovation pipeline. The empirical analysis then links these outcomes to Medicaid demand exposure and PDL exposure, allowing the response to be studied both across therapeutic classes and across firms with different exposure to Medicaid formulary institutions.

The closest related paper is Garthwaite et al. (2021), which studies clinical trial activity around the Medicaid expansion and finds little evidence of an aggregate innovation response. This thesis builds on that work by extending the outcome set to patents and FDA approvals, extending the post-expansion window to 2025, and moving from therapeutic-class-level analysis to a firm-class panel. The empirical results are then used to inform the illustrative model developed later in the thesis, which studies how the Medicaid expansion affects firm innovation incentives when the preferential status on the PDL changes the surplus firms expect to capture.

1.2 Motivation and Policy Relevance

Public programmes are increasingly central to the pharmaceutical market in the United States. A large share of prescription drug spending is financed by the public sector, and recent reforms have placed greater emphasis on using public purchasing power to reduce drug prices. The 2022 Inflation Reduction Act is a clear example of this shift, where Medicare price negotiation have been introduced for high-spend drugs. These reforms raise the same broad policy tension studied in this thesis: lower prices can benefit patients and taxpayers in the short run, but they may also affect the expected returns to future pharmaceutical innovation.

The Medicaid expansion provides a useful benchmark for this wider policy question by combining a large coverage expansion with an established system of public reimbursement and rebate negotiation. Similar questions arise wherever public buyers combine broad coverage with price control, including the NHS in the United Kingdom and other publicly financed health systems.

The implication is not that public bargaining is necessarily harmful. Rather, the point is that public reimbursement rules can affect pharmaceutical innovation incentives, not only current drug spending. This thesis studies that relationship in the context of Medicaid, where public demand expanded while the institutions governing public reimbursement became more important.

The remainder of the thesis is structured as follows. Section 2 reviews the related literature on Medicaid and the ACA, public procurement and pharmaceutical pricing, market size and pharmaceutical innovation, and dynamic innovation incentives. Section 3 provides the institutional background needed for the empirical analysis, setting out the ACA Medicaid expansion, the Medicaid rebate system, and the Preferred Drug List mechanism in greater detail. Section 4 describes the data sources and the construction of the main variables, including Medicaid demand exposure and PDL exposure at the firm-by-class level. Section 5 sets out the empirical strategy for the three research questions. Section 6 reports the empirical results. Section 7 presents the theoretical model, including the structure of the game, the mapping from theoretical states to empirical outcomes, the equilibrium analysis, and the comparative statics. Section 8 discusses the policy implications, focusing on the tension between lower net drug prices and the innovation incentives required to justify upfront R&D investment. Section 9 discusses extensions and directions for further research with section 10 concluding the thesis.

2 The Affordable Care Act

Medicaid is a federal-state public health insurance programme for low-income individuals and other eligible groups in the United States. Prior to 2014 eligibility was largely restricted to specific categories, including low-income children, pregnant women, disabled individuals, and some elderly individuals, as opposed to low-income adults in general. While prescription drug benefits were formally non-mandatory, all state Medicaid programmes provided them. Additionally, outpatient prescriptions are reimbursed through fee-for-service and managed care arrangements.

The ACA substantially expanded this market. From 2014, states were allowed to extend Medicaid eligibility to adults with incomes up to 138% of the federal poverty line. This threshold change generated a large increase in Medicaid enrolment, and therefore in the number of prescriptions likely to be reimbursed through Medicaid.

Importantly, there are two key institutional features of Medicaid that are noteworthy. The first is the Medicaid Drug Rebate Program (MDRP). Manufacturers that want their drugs to be reimbursed by Medicaid must enter into a national rebate agreement and pay rebates to states for each unit dispensed to Medicaid. Specifically, for brand-name drugs, the basic rebate applied is based on what is greater between a fixed percentage of the average manufacturer price (AMP), or the difference between AMP and the manufacturer's lowest commercial price, known as the "best price". Consequently, discounts offered elsewhere in the market can affect the rebates owed to Medicaid (Duggan and Scott Morton, 2006). Medicaid therefore receives a legally protected discount thereby lowering the value of Medicaid prescriptions to manufacturers.

The ACA also changed the rebate environment directly. The statutory rebate percentage under the MDRP was increased from 15.1% to 23.1% of AMP for most brand-name drugs, and from 11% to 13% for generics. Firms were now obligated to extend rebates to drugs dispensed through Medicaid managed care organisations, while previously being exempt.

The second institutional feature is the Preferred Drug List (PDL). States can negotiate supplemental rebates from manufacturers in exchange for preferred placement on their formulary. Preferred drugs are prescribed to patients without prior authorisation, while non-preferred drugs require additional approval before the prescription is filled. Prior authorisation creates an administrative cost for physicians, therefore preferred placement can affect the prescribing behaviour of a physician and the Medicaid market shares within a therapeutic class as a result. Preferred placement therefore gives states a source of negotiating leverage. Since preferred status affects access conditions and expected utilisation within a therapeutic class, manufacturers may offer supplemental rebates above the statutory MDRP minimum in order to obtain or retain preferred placement. Additionally, some states also participate in multi-state purchasing alliances, which aggregate Medicaid demand across states and may further strengthen their bargaining position.

Selection into the PDL takes on the form of a two-stage selection mechanism. In the first stage, the state's Pharmacy and Therapeutics committee assesses whether a drug is clinically appropriate for preferred status. This involves the use of clinical evidence from peer-reviewed clinical literature and oncology reports to understand the safety, efficacy and effectiveness of the drug relative to to other drugs with the same role. In the second stage, once drugs pass the first stage and are concluded to be therapeutically comparable, states consider the net cost and supplemental rebates when deciding which drugs receive preferred placement. This setting matters as it means that Medicaid procurement does not only depend on clinical value, but also on firms' willingness to offer price concessions. For this reason, an expansion of Medicaid demand may also increase the value of preferred placement and intensify competition over rebates.

Importantly, preferred placement on the PDL is not necessarily limited to a single drug for a given drug role. States may allow for multiple preferred drugs within the same role in order to allow for clinically appropriate choices among patients, such as int the case of route of administration, prior treatment response or clinical needs such as allergies. Furthermore, drugs with non-preferred status are not necessarily from Medicaid coverage, they would instead require authorisation from Medicaid once a physician requests prescribing it.

Thus, the ACA created a demand shock, but it did so inside a public procurement system with rebates and formulary bargaining. For high-Medicaid therapeutic classes, the reform therefore increased the number of prescriptions likely to be paid for by Medicaid, while also increasing the importance of the rebate and PDL mechanisms that determine the net revenue firms receive from those prescriptions.

This thesis does not utilise the staggered timing of state adoption as the main source of variation. Although that variation is useful for studying outcomes such as insurance coverage, healthcare use, or labour supply, it is less suited towards studying pharmaceutical innovation. Pharmaceutical innovation is a national investment decision, and firms cannot develop a drug for sale in one expansion state alone. Instead, they consider expected sales across the United States. Therefore, this thesis treats the start of the ACA Medicaid expansion in 2014 as the relevant policy shock.

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